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const axios = require("axios");
const fs = require("fs");
const path = require("path");
const OpenAI = require("openai");
const config = require("../config");

const openai = new OpenAI({ apiKey: config.OPENAI_API_KEY });

async function generateDesignConcept(theme) {
  const completion = await openai.chat.completions.create({
    model: "gpt-4",
    messages: [
      {
        role: "system",
        content: "You are a digital product designer for Etsy. Generate image pro",
      },
      {
        role: "user",
        content: `Create a design concept for this theme: "${theme}"

Return ONLY this JSON:
{
  "imagePrompt": "Detailed Stability AI prompt. Flat design, vector style, white
  "productType": "e.g. Clipart Set, Wall Art, Printable, SVG Design",
  "fileName": "snake_case_file_name_no_spaces"
}`,
      },
    ],
  });

  const raw = completion.choices[0].message.content.trim();
  return JSON.parse(raw.replace(/```json|```/g, "").trim());
}

async function generateImage(prompt, outputDir) {
  const response = await axios.post(
    "https://api.stability.ai/v2beta/stable-image/generate/core",
    {
      prompt,
      output_format: "png",
      width: config.DESIGN_WIDTH,
      height: config.DESIGN_HEIGHT,
      style_preset: "digital-art",
    },
  ),

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    {
      headers: {
        Authorization: `Bearer ${config.STABILITY_API_KEY}`,
        Accept: "image/*",
        "Content-Type": "application/json",
      },
      responseType: "arraybuffer",
    }
  );

  if (!fs.existsSync(outputDir)) fs.mkdirSync(outputDir, { recursive: true });
  const filePath = path.join(outputDir, "design.png");
  fs.writeFileSync(filePath, Buffer.from(response.data));
  return filePath;
}

async function generateDesign(theme, outputDir) {
  const concept = await generateDesignConcept(theme);
  const designPath = await generateImage(concept.imagePrompt, outputDir);
  return { ...concept, designPath };
}

module.exports = { generateDesign };

```